

Breeding

Birds lay their eggs either directly on the ground or in a nest. One parent – or both – keep the eggs warm by sitting on them, or incubating them. Young birds hatch from eggs at different stages of development. Some can look after themselves almost immediately; others rely on their parents for food and protection.



Eggs

Birds' eggs have a hard shell. Ground-nesting birds often lay eggs that match their background. Birds that nest in trees often lay plain eggs.



Helpless young

Tree-nesting birds usually produce poorly developed young without feathers. The young stay in the nest until they are ready to feed themselves.



Well-developed young

The young of most ground-nesting birds can feed within hours of hatching. They soon leave the nest and follow their mother.

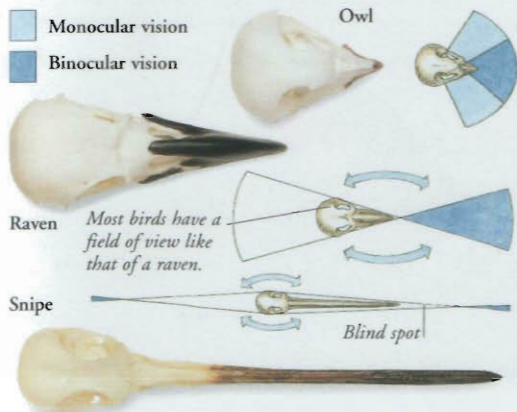


Foster parents

Brood parasites are birds that trick others into raising their young. Here a reed warbler is feeding a cuckoo that hatched in its nest.

Monocular vision

Binocular vision



Vision

Birds that hunt, such as owls, have eyes at the front. This restricts their field of view, but they can judge distances accurately. Shorebirds, such as the snipe, have eyes at the side. They can spot danger in any direction, including behind.

Senses

For most birds, vision is by far the most important sense. It guides them to their food and helps them to avoid their many enemies. Hearing plays a part in helping birds to communicate, and is important to birds that hunt in the dark. The sense of smell is far less vital to birds than it is to many other animals, although some birds, such as the kiwi, use it to find food.

Crane

Like most birds, a crowned crane has keen eyesight. Its eyes are so big that they almost meet in the centre of the skull. Its ear openings are at the base of its crown, but they are hidden by short feathers. Its nostrils are in its beak.



Crowned crane

Flight

This complex way of moving requires superb co-ordination. Some birds stay airborne almost entirely by flapping, but others hold their wings out and glide through the air, using the natural curve of their wings to provide lift. During flight, a bird adjusts the shape of its wings to alter its speed and height.



A pigeon's wings allow good manoeuvrability when extended, and fast flight when partly closed.



A kestrel's large wings provide lift as the bird flaps them non-stop while it hovers in the air.



A grouse's wings are shaped for load-bearing rather than speed. A grouse flies only in short bursts.



A peregrine falcon's slender wings partly fold up when it dives out of the sky on to its prey.

Wing shapes

Birds have evolved a variety of wing shapes that enable them to fly in different ways. Some wings provide lots of lift but do not work well at speed. Others create as little friction as possible when they cut through the air, allowing a bird to fly faster.

Flightless birds

During the course of evolution, some birds have given up being able to fly. Flightless birds do not need a light body, and although some are quite small, they include the biggest birds that have ever lived.

The flightless rhea comes from South America.



Feeding and diet

Birds spend much of the time looking for food. To be able to fly, birds need food that provides them with lots of energy. Many of them eat small animals, which they catch either on land, in the air, or in water. Others visit plants and eat fruits, seeds, nectar, and pollen. Some have a mixed diet. Unlike mammals, only a few birds eat grass or the leaves of other plants.



Fish eaters

The great blue heron catches fish by stabbing them with its beak. Other fish eaters snatch their prey with talons, dive-bomb them from above, or chase them through the water.



Seed eaters

Different birds eat different seeds. They usually crack open the seed's husk before eating the food inside. The goldfinch is a typical seed eater. It feeds on thistles.



Insect eaters

Insect-eating birds search for their food on the ground or on plants, or snap it up in mid-air. The goldcrest often feeds high up in trees. Like other small insect eaters, it is expert at spotting insects hidden on leaves or bark.

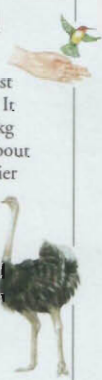


Meat eaters

Many birds eat small animals, but owls and birds of prey specialize in hunting larger animals, such as mammals, reptiles, and other birds. A hooked beak allows them to tear up their food before swallowing it.

Largest and smallest

The world's heaviest bird is the ostrich. It weighs up to 125 kg (275 lb). This is about 80,000 times heavier than the rare bee hummingbird, the smallest bird. This tiny bird's eggs are the size of peas.



FIND OUT MORE

ANIMAL BEHAVIOUR

BIRDS OF PREY

EGGS

FLIGHT, ANIMAL

FLIGHTLESS BIRDS

NESTS AND BURROWS

OWLS AND NIGHTJARS

SKELETON

SONGBIRDS